



Fate of the Caribou Project

2024 NSF Annual Report

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ACCOMPLISHMENTS

What are the major goals of the project?

Caribou (*Rangifer tarandus*) are the most widespread large mammal of the Arctic and therefore both an ecological keystone species, and of profound importance to the culture and lifeways of Indigenous and settler peoples of the north. Climate change is rapid in the Arctic and a warmer climate threatens the ecological, cultural, and economic role of caribou. The Fate of the Caribou project's goal is to advance understanding of the coupled human-natural system of the Arctic to inform the adaptations of our collaborating communities to climate-induced changes in caribou distributions. Led by the concerns and observations of local communities and Indigenous governing bodies and using shared datasets that are unprecedented in scale and scope, we are implementing a large-scale, comparative suite of analyses across multiple caribou herds in North America, developing novel ecological modeling and environmental sensing tools at a near-continental scale.

Our objectives are to:

- O1. Build on new and existing partnerships with local communities and Indigenous, state, territorial, and federal governments in Alaska and Canada to share data and tools, and to guide research questions.
- O2. Identify continental-scale and herd-specific processes that control the dynamics and distributions of caribou populations.
 - O2a. Link environmental change, especially human development, directly to demographic processes and population dynamics using caribou movement and behavioral mechanisms;
 - O2b. Assemble and synthesize spatially-explicit environmental layers of vegetation change, fire dynamics, snow quality, and human infrastructure;
 - O2c. Develop dynamic field, remote sensing, and community-derived environmental datasets at a near-continental scale to measure the effects of human development and climate change on caribou and their habitat;
 - O2d. Develop analysis tools, data visualizations, and storytelling products that can be used by resource management organizations and stakeholder and scientific communities to share the dynamics, distributions, and importance of caribou and to inform policy;
 - O2e. Make significant methodological advances in ecological, behavioral, habitat, and population modeling and in frameworks for the co-production of knowledge with Indigenous partners.
- O3. Support three postdoctoral researchers, three PhD students, and several undergraduate technicians in the fields of wildlife ecology, environmental informatics, and natural resource management.

What was accomplished under these goals and objectives?

Major Activities: Working with our established partners, we conducted data collection, analysis, and synthesis on all major research questions, while also exploring new areas of interest and concern brought

up by collaborators. This included development and deployment of new caribou audio recording equipment, initiation of a collaborative camera-trap survey, refinement of remote-sensing methods to detect snow and ice, and more. Half of our team traveled to communities in Alaska and northern Canada throughout this year to participate in research and policy meetings, meet with community members, and deploy research equipment. Additionally, we published and improved several ground-breaking and publicly-available datasets and analysis tools. We shared our progress and findings through multiple communication pathways, including scientific papers, graphical summaries, reports, multi-media essays, and community presentations.

Specific Objectives:

- O1. Building on new and existing partnerships:** We have strengthened and expanded our partnerships with Indigenous groups this year, particularly Tłı̨chǫ Government, Wek'èezhìi Renewable Resources Board, North Slave Métis Alliance (NSMA), and Kugluktuk Angoniatit Association. Team members conducted focus groups with NSMA members, hearing about changes in caribou populations and migration patterns, hunting and cultural practices, and their concerns for caribou. Participants worked with our team to (a) identify the community's concerns about caribou to guide research; (b) develop collaborative relationships for ongoing review of methods, results and deliverables; and (c) find the best ways to communicate efforts and involve NSMA members in the research. Post-doc Couriot was also invited to join NSMA's monitoring of caribou and traffic activity on the Tibbit to Contwoyto Winter Road, assisting with observations and giving feedback on fieldwork protocol. PhD student Perra worked with NSMA to spatially analyze traffic noise based on NSMA's array of acoustic recorders along several mine roads.

Team members presented at the 2023 Western Arctic Caribou Herd Working Group meeting in Anchorage, Alaska, leading to discussion on caribou survival and movement. The Working Group – which consists of representatives from 15 communities in northwest Alaska, and 5 non-Indigenous interests – is the respected co-management authority on the Western Arctic herd and has a direct impact on policy. This partnership continues to develop, and we will again be attending the Working Group meeting this winter.

- O2a. Linking environmental change and caribou:** We continue to use GPS collar data to explore drivers of adult caribou survival and migration in a dynamic and heterogeneous environment. We have a paper in press reporting that large-scale behavioral shifts in migratory caribou in Alaska may, in part, reflect strategies to mitigate poor survival under climate change. Survival and demographics are the focus of PhD student Beaupré's work, which also involves a reassessment of boreal caribou demography and impacts of disturbance in the Northwest Territories. This year, Beaupré has expanded her analysis of boreal caribou survival and fostered research partnerships with Government of Northwest Territories, Environment Canada, and Natural Resources Canada, culminating in organizing and attending the Boreal Caribou Survival Project Workshop.

Further analysis is underway to build a synthetic population model for caribou. PhD student McConnell has built the foundational models and is exploring ways to incorporate stochasticity and complex environmental variables. Following setbacks with data sharing agreements this year, McConnell pivoted to develop a theoretical model, allowing for flexible exploration of biological and environmental covariates.

PhD candidate Liao has continued analysis of caribou movement and lake freeze and thaw events. Without lake ice, caribou must swim across or circumnavigate large lakes, both of which are energetically costly. Liao is refining remote sensing tools to detect lake ice at fine spatial and temporal scales, then comparing ice status with caribou movement data. Liao successfully defended her project proposal this year and is now a PhD candidate.

Post-doc Larue is working with the Kugluktuk Angoniatit Association (KAA), whose members are concerned that grizzly bears may be increasing on caribou calving grounds, which could have detrimental effects on the imperiled Bathurst caribou population. Larue's initial analysis of camera trap data showed that grizzlies and caribou do overlap on the calving grounds, as was hypothesized by KAA members.

- O2b. Assembling spatial datasets:** Our team at Northern Arizona University published their groundbreaking Arctic Plant Aboveground Biomass Synthesis Dataset. The dataset synthesizes plant and lichen biomass data from 32 individual datasets comprising 2327 sample plots across seven Arctic countries. The dataset is available through the Arctic Data Center and constitutes an invaluable resource for the Arctic research community.
- O2c. Developing continental-scale datasets:** Our team is working to consolidate traditional knowledge and local observations that have been shared at caribou management meetings, public hearings, and research interviews into a searchable database ("Knowledge of the Caribou database"). The goal is to provide the technical skills and human power to make existing records of caribou observations accessible to alleviate the information requests to Indigenous partners. Database development is well underway, and we have a working prototype that has been shared with our partners. We are troubleshooting the data entry process and plan to solicit feedback from our partners in the coming year.
- O2d. Analysis tools, data visualizations, and communication products:** Post-doc Couriot built a protocol for identifying caribou calving grounds, which are ecologically critical for caribou. Couriot's approach uses past definitions combined with behavioral analysis tools developed by our team to identify core calving grounds and areas used to raise calves while they are vulnerable.

The TuktuTools R package, debuted last year, is being used by caribou researchers in the Northwest Territories and elsewhere. We have worked with our research partners this year to debug and improve the package and analysis tools. We are also working on a new package, TuktuMigration, which uses methods developed by PI Gurarie and others to estimate migration timing for barren-ground caribou. Post-doc Barbour is building a "Permeability" R package, with the goal of providing tools to estimate the barrier effects of natural and anthropogenic landscape features (i.e., roads, rivers) on migratory species.

At the request of the Tłıchq Government, we are creating an animation of caribou movements around mines and mining roads in the Bathurst Caribou Herd's range, which will help them demonstrate how caribou respond to roads and traffic. This animation is under development, in collaboration with the Wyoming Migration Initiative, Wek'èezhì Renewable Resources Board, and North Slave Métis Alliance.

- O2e. Methodological advances:** Our team is using acoustic recording and machine learning tools to investigate Arctic acoustic ecology, especially with respect to caribou. PhD student Perra has

made significant strides in building and testing animal-borne acoustic recorders, successfully recording and documenting different behaviors on captive reindeer. Deployment of recorders on caribou collars in Northwest Territories faced setbacks, including equipment failures that made retrieval of the recorders difficult. However, Perra continues to make modifications to the recorders so we can learn what the caribou hear, especially close to mine sites.

Gurarie, Couriot, Beaupré, and Barbour continue work on new analysis methods based on cognitive movement ecology (how choice and collective memory drive caribou movement and survival), leading to new insights into a range shift in the Western Arctic Caribou Herd and development of tools to delineate calving areas. Gurarie led a class of graduate students and post-docs in exploring applications and limitations of the analysis tools.

- O3. Supporting early career scientists:** Our project is fully staffed, including four post-doctoral researchers (one of whom is now a professor at the University of Alaska), five PhD students/candidates, a program and communications manager, and several undergraduate research associates. We are also recruiting for another graduate student position to lead development and analysis on the Knowledge of the Caribou database project.

Significant Results:

- O2a. Linking environmental change and caribou:** PI Gurarie and Post-doctoral researcher (now professor) Couriot and PhD student Beaupré completed a novel analysis of large-scale caribou range shifts in the Western Arctic Caribou Herd and found the shifts were driven by social and cognitive mechanisms to process and act on environmental changes (Gurarie et al. *in press*). This is a significant result as it points to fundamental mechanisms of adaptation to climate change for a highly-mobile organism.

With colleagues at Lehigh University in Pennsylvania, PI Gurarie contributed to an important undergraduate-led study (Matias et al. 2024) showing that one of the strongest triggers for caribou migrations is snow *quality* rather than snow *quantity*, in particular the moment at which snow begins a daily cycle between melting and freezing.

PhD candidate Qianru Liao successfully used caribou tracking collar data to classify instances of caribou crossing lakes and rivers or circumnavigating them during migration. This fine-scale behavioral classification would have been impossible a few years ago, but improved tracking equipment and analysis allow Liao to infer each movement “step” caribou take during migration. She further refined use of satellite imagery to detect snow and ice based on the amount of light reflecting off the earth’s surface (albedo data) and has built preliminary models tying the presence of snow and ice to caribou crossing or circumnavigating lakes and rivers. Ice conditions impact water-crossing behavior and that in turn affects caribou energy levels and their ability to give birth safely and successfully. Liao is using her combined caribou and albedo datasets to investigate how caribou mobility is affected by changing freeze/thaw conditions, and whether caribou are adapting their migratory behavior to adjust to those conditions.

Following discussion with members of the Kugluktuk Angoniatit Association (KAA) in Nunavut, Canada, Post-doctoral researcher Larue conducted analysis on camera trap data collected by KAA on the calving grounds of the Bathurst Caribou Herd. His analysis showed that grizzly bears significantly overlap with caribou during the calving season.

Post-doctoral researcher Ehlers used tracking collars equipped with video cameras to link fine-scale foraging ecology to juvenile caribou survival and provide caribou managers critical insight to the density-dependence of an abundant caribou herd (Fortymile Herd). Ehler's results are already informing management decisions (i.e., hunting quotas) of the Fortymile Caribou Herd in eastern Alaska.

In a preliminary study performed in collaboration with the Yukon Government and presented to the Porcupine Caribou Meeting Board and Porcupine Caribou Technical Committee, co-PI Couriot found that the Porcupine Caribou Herd (PCH) has shifted their fall migration to the Yukon, discontinuing use of a migration corridor crossing the Brooks Range in Alaska. The PCH is under intense ecological and political scrutiny as it ranges between Alaska and Canada, and calves in the hotly-contested Arctic National Wildlife Refuge (ANWR). This range shift could have major impacts on communities relying on the PCH for subsistence, and on industrial development in ANWR. Discussions with Indigenous communities in the PCH range are planned for November 2024 and will focus on sharing preliminary results and collaboratively developing hypotheses for changes in the seasonal distribution of the herd. Further analyses will be performed following the discussions.

- O2e. Methodological advances:** PhD student Perra has designed innovative animal-borne acoustic recording devices, using 3D printing to build recorders that are lightweight enough to be attached to caribou tracking collars without negatively affecting the caribou. While the initial deployment of those devices encountered difficulties typical of a pilot study in the Arctic (one 1 of 22 were recovered), they have been redesigned, improved, and are currently being tested for a more robust deployment in 2025.

Additionally, Perra's analysis of audio data from stationary audio recording devices has narrowed down the temperature and wind thresholds necessary for alleviating biting insect harassment on caribou. Biting insects cause blood loss and energy expenditure for caribou in the summer, and the warming climate allows insects to be active for longer periods. Perra can detect insects in audio data and is tying insect activity levels to local temperature and wind data, improving previously-established thresholds for modeling insect harassment. This analysis supports Inuit and Dene knowledge that Contwoyto Lake is critical insect-relief habitat for the threatened Bathurst Caribou Herd. Contwoyto Lake is being considered for several new infrastructure projects, making her results more important than ever.

Post-doctoral researcher Barbour and PI Gurarie are spearheading the development of a *permeability* R package which quantifies and models the extent to which linear elements (especially roads) inhibit the movement of wild animals, with an emphasis on proposed expanded Arctic infrastructure on movements of caribou. This is methodological development with wide-reaching implications, and is of particular interest to Northern communities who have significant concerns regarding the fragmentation of the caribou range.

PI Gurarie with PhD student Beaupré and post-doctoral researchers Couriot and Barbour are building on the analysis of the Western Arctic Caribou Herd range shifts to develop new techniques to analyze animal movements that take sociality and spatial memory into account, as opposed to focusing primarily on environmental drivers. These approaches bring Western scientific approaches to animal movement analysis more in line with traditional and Indigenous

knowledge, which is more aware of the collective movements and long memories of a highly-mobile and highly-social species.

Key outcomes or other achievements:

O2a. Linking environmental change and caribou: Post-doc Libby Ehlers published papers built on a novel dataset from video camera-equipped tracking collars, which allowed researchers to report caribou foraging ecology at an extremely fine scale, identifying forage species down to the number of bites taken. Ehlers and Larue are working on cataloging the video camera data on MoveBank, an online database of animal tracking data used by researchers worldwide.

O2b. Assembling spatial datasets: Our remote sensing team's publication of the Arctic Plant Aboveground Biomass Synthesis Dataset is a significant resource for the entire Arctic research community. When applied to caribou demographic models like those being developed by McConnell and other Fate of the Caribou team members, it will provide insight at a much larger spatial scale than existing datasets.

O2d. Analysis tools, data visualizations, and communication products: Our partnerships with Indigenous governments and caribou co-management boards have led to the development of an online storymap (an interactive storytelling and data visualization platform) outlining our recent movement analysis and Indigenous knowledge about the Bathurst Caribou Herd in northern Canada. After over two years of development with partners from the Wek'èezhii Renewable Resources Board, Tłıchq Government, Government of the Northwest Territories, and the Wyoming Migration Initiative/Global Initiative on Ungulate Migration, the storymap was published in November 2023. It has been shared by our partners via social media and was featured in the North Slave Métis Alliance membership newsletter and Wek'èezhii Renewable Resources Board website. We have also generated several other outreach and communication products this year, including our first research summary in the Tłıchq language.

Post-doctoral researcher Couriot is revising a manuscript detailing her technique for delineating calving and calf-rearing areas. This paper will be a major milestone for caribou conservation and industrial development planning and mitigation. While development projects often report not to impact sensitive calving areas, historic inconsistencies and disagreements about which areas are most important to caribou have convoluted mitigation and conservation efforts. Couriot's method is both ecologically-sound and reproduceable, addressing those historic inconsistencies.

What opportunities for training and professional development has the project provided?

To date, the Fate of the Caribou project has recruited three post-doctoral researchers (at SUNY-ESF, University of Montana, and Northern Arizona University) and five PhD students (two at SUNY-ESF, one at Northern Arizona University, and two at University of Maryland), and have supported paid work for several undergraduate researchers. PI Gurarie is in the process of recruiting an additional graduate student to lead further development and analysis on the Knowledge of the Caribou database project. Post-doctoral researcher Couriot has recently been offered an assistant professor position at the University of Alaska Fairbanks Department of Biology and Wildlife.

We also continue to grow our team's technical skillset through interdisciplinary partnerships. For example, PI Fagan is supporting an undergraduate computer science major who is working on the technical build-out of the Knowledge of the Caribou database and Program Manager Brose is working

with cartographers at the Wyoming Migration Initiative to animate maps of caribou movement around mining roads.

PhD candidate Liao participated in the Applied Stochastic Processes for Encounter Problems in College Park, Maryland to learn new theories and applications of stochastic processes in two-dimensional spaces. She also attended training on estimating the density of unmarked animals from camera trap data and on statistical tools for analyzing continuous-time animal tracking data, both of which will help her improve her ecological modeling skills.

At Northern Arizona University, PhD student Salganek attended the Geo for Good Summit at the Google Earth Engine Conference in Mountain View, California. The meeting brought together scientists, agency personnel, and decision makers interested in the use of technology for environmental and social good.

Several team members gave guest lectures as invited speakers at various universities and institutes, including the University of Alaska Fairbanks, University of Alaska Southeast, Universidade de Mato Grosso do Sul in Brazil, and University of California San Diego.

Post-doctoral researcher Couriot and PhD student Beaupré completed the “Belonging, Accessibility, Justice, Equity, Diversity, and Inclusion” offered by the Polar Science Early Career Community Office and Couriot completed training on lab animal research through the State University of New York.

To highlight, in particular, the contribution of undergraduate students: one (C Wright) is leading a manuscript using camera trap data, collected under the leadership of the Kugluktuk Angoniatit Association, to analyze parasite loads on caribou and female fecundity and health, applying non-invasive methods consistent with Indigenous values. Another (J Morey) is leading the technical development for the Knowledge of the Caribou project. A third affiliated student (M Matias, Lehigh University) was the first author on an analysis of caribou migration and remotely-sensed snow quality.

Have the results been disseminated to communities of interest?

Our progress and results have been communicated primarily through in-person meetings and presentations this year. We have spent considerable time traveling to our collaborating communities, including Yellowknife, Northwest Territories, and surrounding communities to meet with partners and Indigenous knowledge holders. The focus of these trips was to listen and learn, rather than disseminate results: we spent extensive time with caribou specialists and local knowledge holders out “on the land”, learning about the ways wildfire is changing vegetation regimes, how caribou are responding to mining infrastructure, and how Indigenous groups are leading the way in caribou research, monitoring, and protection through their Caribou Guardians programs. During those trips, we shared information about our work and technical skills as a tool for collaborative research to address concerns and questions posed by the people who are living alongside the caribou we study.

In several cases during travel to northern Canada, we presented data visualizations and preliminary results to community members and caribou advisory groups to demonstrate our in-progress work and facilitate feedback. Most notably, Couriot, Perra, Beaupré, and Gurarie conducted several focus groups with members of the North Slave Métis Alliance to better understand members’ perspective and concerns for caribou, and to identify best practices for collaborating on research and sharing results.

We presented our research at numerous conferences, workshops and community meetings this year, including at the NNA Community Meeting, International Statistical Conference, The Wildlife Society Annual Meeting, and The Alaska Chapter of the Wildlife Society Annual Meeting. We gave focused research updates to partners at Government of Northwest Territories (GNWT), presentations and discussion at the Western Arctic Caribou Herd Working Group meeting in Alaska and participated in the Collaborative Barren-ground Caribou Initiative Year One Workshop hosted by the GNWT Cumulative Impacts Monitoring Program and Polar Knowledge Canada. All three of these meetings included researchers, stakeholders, and policy makers directly impacted by and impacting the caribou herds we study. PhD student Beaupré helped organize the Boreal Caribou Survival Workshop, which brought together researchers and policy makers in Northwest Territories to discuss ongoing research and preliminary results on the survival of imperiled Boreal woodland caribou in the Northwest Territories. Co-PI Gunn served on the Wek'èezhii Renewable Resources Board technical committee and attended Technical Advisory Committee meetings for mining operations on behalf of the Kivalliq Inuit Association, including sharing relevant results and tools from the Fate of the Caribou project.

We published several communication and outreach products that have been under construction, including “Shifting Trails: the shrinking range of Bathurst Caribou”, a multimedia experience (ArcGIS Storymap) telling the story of the rapid population decline and subsequent range shifts of the Bathurst Caribou Herd (shiftingtrails.caribou.esf.edu). This project was a major collaborative effort between our team, the Tłı̨chʼı̨ Government, Wek'èezhii Renewable Resources Board (WRRB), Wyoming Migration Initiative, and other partners. It combines our spatial analysis of the Bathurst caribou range with the cultural context and real-life impacts experienced by Tłı̨chʼı̨ communities who are no longer seeing caribou where they used to. The essay has been shared via social media by the WRRB and via email to the entire membership of the North Slave Métis Alliance (<https://www.nsma.net/post/march-newsletter>), who also live in the Bathurst caribou range.

Program and Communications Manager Brose led design and revision of “Are Bathurst caribou overlapping more with other herds in winter?” (https://www.wrrb.ca/sites/default/files/BathurstOverlap_digital.pdf), a one-page graphical summary of research completed by Gurarie, Couriot, and Gunn in partnership with the WRRB. Winter range overlap between the protected Bathurst herd and neighboring herds has made it hard for community members to safely and efficiently harvest legal caribou from the neighboring herds, so this research is of major interest to communities in the Bathurst range. We worked with a translator to produce a Tłı̨chʼı̨ language version to increase accessibility and reach (https://www.wrrb.ca/sites/default/files/WinterOverlap_Poster_Tlicho_digital.pdf). The graphic has been shared via social media by the WRRB and via email to NSMA members (<https://www.nsma.net/post/caribou-research-underway-update>). Brose is working on a similar graphic for research recently co-authored by Gurarie on the effects of snow and ice conditions on caribou migration.

We discontinued our Twitter (X) account this year as X's policies and culture no longer aligned with our project objectives and values. We also recognized that our posts on X were primarily reaching other researchers, which is important, but we decided efforts were better spent trying to reach the people most impacted by caribou. Following guidance from Navigating the New Arctic's 2022 Broader Impacts Network training on social media and feedback from our research partners, we launched a Facebook page in February 2024 (<https://www.facebook.com/FateOfCaribou>). Growth on the page has been slow, but we are slowly building an online community, sharing our work, and engaging with people interested in caribou.

Our work was featured in two Arctic community-focused publications this year. Brose and others authored an illustrated essay that was published in the Xàgots'eèhk'ò Journal (<https://xagotseejournal.com/index.php/xgsk/article/view/5966>), a journal “for Northerners, by Northerners”, based in Northwest Territories, Canada. The essay was featured in a special issue on caribou alongside art, poems, and recipes from community members. Brose also wrote a project highlight for the Summer 2024 “Caribou Trails” newsletter produced by the Western Arctic Caribou Herd Working Group. The newsletter is printed and mailed to all residents in the Western Arctic Caribou Herd’s range – an estimated 10,000 people, in addition to being available online (https://westernarcticcaribou.net/wp-content/uploads/2024/08/CaribouTrails_2024_508-hi-res.pdf). We also have several submissions to the upcoming issue of NNA’s Nuna Zine, featuring an interview with post-doc Larue and poems by Beaupré and Brose that put a creative spin on caribou behavior and migration.

Our research efforts were featured in several online articles and blog posts, including a feature by University of Maryland’s College of Computer, Mathematical, and Natural Sciences (<https://cmns.umd.edu/news-events/news/bill-fagan-climate-changes-impact-arctic-caribou>), a LinkedIn post from the same (<https://www.linkedin.com/feed/update/urn:li:activity:7193947248152453120/>) an article on the Arctic Plant Aboveground Biomass Synthesis Dataset (<https://arcticdata.io/all/news/2024/06/new-dataset-reveals-insights-into-arctic-plant-biomass-with-dr-logan-berner/>), and a blog post on muskox genetic research by Co-PI Gunn (<https://communities.springernature.com/posts/genome-assembly-of-a-canadian-arctic-island-banks-island-muskox-encouraging-news-for-conservation>).

Post-doc Couriot led efforts on several reports for our research partners, including the North Slave Métis Alliance and Government of Yukon. Co-PI Gunn also drew on Fate of the Caribou results to author several reports for the National Oceanic and Atmospheric Administration’s Arctic Report Card, the World Wildlife Fund Global Arctic Programme, and Government of Northwest Territories. These types of reports are just as impactful as peer-reviewed papers because they are written for policy makers and are directly referenced in decision-making settings.

What do you plan to do during the next reporting period to accomplish the goals?

At SUNY-ESF, PhD student Megan Perra will continue her research on the acoustic ecology of barren-ground caribou, especially in relation to human development and infrastructure, which is a key concern for our Indigenous partners. This involves continued development of a novel animal-borne acoustic recorder and deploying both animal-borne and stationary acoustic recorders in northern Canada. PhD student Beaupré is working on analysis of caribou survival and adaptations related to human infrastructure and climate change. Perra plans to defend her proposal and both Perra and Beaupré will do qualifying exams and advance to PhD candidacy this year. PI Gurarie and post-docs Couriot and Barbour will continue developing behavioral analysis tools including improvements on the TuktTools, TuktMigration, and Permeability R packages. Couriot will be focusing on the Porcupine Caribou Herd in partnership with Government of Yukon, while Gurarie and Barbour continue various analyses on other barren-ground and boreal woodland caribou in Canada and Alaska.

At the University of Maryland, PhD student McConnell will finalize analyses and write the first draft of a manuscript on caribou complex population dynamics by end of 2024, get feedback from collaborators and revise the manuscript for submission in early 2025. PhD student Liao will complete two manuscripts focusing on caribou water crossing behavior, one focusing on caribou movements in the immediate

vicinity of Contwoyto Lake (crossing vs. circumventing) and the other on perception / spatial decision making.

At the University of Montana, Post-doc Larue will co-lead with Indigenous partners the publication of a manuscript on the predator-prey spatiotemporal dynamics of grizzlies and caribou on the calving grounds of the Bathurst herd in the Northwest Territories and Nunavut. This study will help inform communities on the mostly unknown role of grizzlies on caribou survival. This study will thus potentially be critical for predator control which has historically been mostly focused on wolves. The Montana team plans to publish and publicly archive the GPS video-collar data from the Fortymile caribou herd on MoveBank and Scientific Data. Larue will also lead the publication of a methods paper validating the accuracy of Hidden Markov Models for inferring behavioral states from GPS-collar location data. Results from this study will potentially open the door to wide-use of Hidden Markov Models to infer behavior and the use of video-collars for validation. Following the successful collaboration to document grizzly activity on caribou calving grounds, Larue will lead fieldwork to expand his study of grizzlies to the Bluenose East Caribou Herd calving grounds, complete outreach about the project, and present initial results at the ArcticNet Arctic Change conference. He also plans to publish an analysis on the effects of wind speed and direction on caribou movement patterns over periods of insect harassment during summer, and perform data gathering and initial analysis on a broad-scale caribou-road ecology project.

Co-PI Gunn plans to publish a manuscript integrating climate changes with the energetics and demography of migratory tundra caribou, and a manuscript on the implications of overlapping distribution among neighboring caribou herds. She will also continue her advisory role for other Fate of the Caribou project efforts, and continue serving on various technical and advisory committees in Canada.

At Northern Arizona University, the team plans to publish a manuscript analyzing changes in density and range distribution of the Fortymile caribou herd and linking caribou distribution to changes in vegetative cover. Post-doc Orndahl will lead publication of a manuscript on the Pan-Arctic Vegetation Biomass Mapping project, and archiving the Pan-Arctic Vegetation Maps on the Arctic Data Center. They plan to perform outreach activities related to the newly-developed Pan-Arctic vegetation biomass maps including presenting the results at scientific conferences, media interviews, invited articles, and blog and social media posts. The team also plans to publish manuscripts quantifying and analyzing changes in Pan-Arctic vegetation biomass over past two decades and on resource selection by the Western Arctic Caribou Herd during winter. They will also archive their SnowModel simulation data on the Arctic Data Center.

A primary objective for the upcoming reporting period is to continue engaging directly with caribou co-management boards, including once again presenting at the Western Arctic Caribou Herd Working Group annual meeting in Anchorage, Alaska. Program and Communications Manager Brose plans to continue development of graphic summaries of our research to better disseminate results to non-scientific audiences, and we will continue building our online presence via our website and Facebook page.

Various segments of our teams will travel to Alaska and northern Canada in the coming year to participate in research updates, conduct field work, and continue building relationships with communities.

PRODUCTS

Books:

E. Gurarie, N. Barbour, O. Couriot (in development). Techniques and Concepts in Movement Ecology with applications in R. Open-source eBook. <https://eligurarie.github.io/MovementEcologyBook/>

Book Chapters: None

Inventions: None

Journals or Juried Conference Papers:

Berner, L. T., K. M. Orndahl, M. Rose, M. Tamstorf, M. F. Arndal, H. D. Alexander, E. R. Humphreys, M.M. Loranty, S. M. Ludwig, J. Nyman, S. Juutinen, M. Aurela, K. Happonen, J. Mikola, M. C. Mack, M. R. Vankoughnett, C. M. Iversen, V. G. Salmon, D. Yang, J. Kumar, P. Grogan, R. K. Danby, N. A. Scott, J. Olofsson, M. B. Siewert, L. Deschamps, E. Lévesque, V. Maire, A. Morneau, G. Gauthier, C. Gignac, S. Boudreau, A. Gaspard, A. Kholodov, M. S. Bret-Harte, H. E. Greaves, D. Walker, F. M. Gregory, A. Michelsen, T. Kumpula, M. Villoslada, H. Yläne, M. Luoto, T. Virtanen, B. C. Forbes, N. Hölzel, H. Epstein, R. J. Heim, A. Bunn, R. M. Holmes, J. K. Y. Hung, S. M. Natali, A.-M. Virkkala, and S. J. Goetz. 2024. The Arctic Plant Aboveground Biomass Synthesis Dataset. *Scientific Data* 11(1), 305. doi:10.1038/s41597-024-03139-w. NSF support acknowledged. Audience: Arctic research community.

Brose, A., Perra, M., Gunn, A., Gurarie, E., and Beaupré, C. 2024. Fate of the Caribou: Studying caribou and climate, with communities. *Xàgots'eèhk'ò Journal*, 2(2). <https://xagotseejournal.com/index.php/xgsk/article/view/5966>. NSF support not acknowledged. Audience: Residents of northern Canada.

Clark-Wolf, T. J., et al. Caribou and reindeer population cycles are driven by top-down and bottom-up mechanisms across space and time. *Ecology & Evolution*. In Revision. NSF support acknowledged. Audience: caribou researchers.

Couriot, O., Gurarie, E., Cosby, O., Leblond, M., and Campbell, M. What is a calving ground? Matching modern tools with ecological definitions. *Journal of Wildlife Management*. In revision. Audience: caribou researchers, policy makers, and stakeholders.

Druckenmiller, M. L., R. L. Thoman, T. A. Moon, L. M. Andreassen, T. J. Ballinger, L. T. Berner, et al. 2024. The Arctic. *Bulletin of the American Meteorological Society* 105:S277-S330. NSF not support acknowledged. Audience: remote sensing and snow and ice researchers.

Ehlers, L.P.W., Bentzen, T., Gross, J., Wells, J., Sutor, M., Herriges, J., Joly, K., Palm, E. and Hebblewhite, M. In revision at *Wildlife Biology (Nordic Society Oikos)*. Linking foraging ecology to juvenile survival for a subarctic ungulate. NSF support acknowledged. Audience: caribou researchers and policy makers.

Ehlers, L., Palm, E., Herriges, J., Bentzen, T., Sutor, M., Joly, K., Millspaugh, J., Donnelly, P., Gross, J., Wells, J., Larue, B., & Hebblewhite, M. (2024). A taste of space: Remote animal observations and discrete-choice models provide new insights into foraging and density dynamics for a large subarctic herbivore. *Journal of Animal Ecology*, 00, 1–15. <https://doi.org/10.1111/1365-2656.14109>. NSF funding acknowledged. Audience: caribou researchers and policy makers.

Gunn, A., Russell, D., Joly, K., Manzo, L., Pellissey, J., Tulagak, J., Whiting, A.J. 2024. Migratory tundra caribou in a warmer climate. In Arctic Report Card 2024 <https://www.arctic.noaa.gov/Report-Card>. In revision. NSF support not acknowledged. Audience: Policy makers responsible for Arctic policy.

Gurarie, E. and Avgar, T. 2024. Editorial: Cognitive Movement Ecology. *Frontiers in Ecology and Evolution*, 12, (IF: 3). <https://doi.org/10.3389/fevo.2024.1360427>. NSF support not acknowledged. Audience: Theoretical and applied ecologists.

Gurarie, E., Beaupré, C., Couriot, O., Cameron, M., Fagan, W., and Joly, K. Evidence for an adaptive, large-scale range shift in a long-distance terrestrial migrant. *Global Change Biology*. In press. Audience: global change ecologists and caribou researchers.

Lok, S., Lau, T.N.H., Trost, B. et al. Chromosomal-level reference genome assembly of muskox (*Ovibos moschatus*) from Banks Island in the Canadian Arctic, a resource for conservation genomics. *Sci Rep* 14, 21023 (2024). <https://doi.org/10.1038/s41598-024-67270-9>. NSF support not acknowledged. Audience: Arctic ungulate researchers and managers.

Kölzsch, A., Hahn, C., Armfield, M., Beaupré, C., Böck M., Brown B., Eichler C., Jain V., Ketchin M., Long J., Merdian-Tarko A., Minchin L., Neves T., Rakesh A., Safi, K., and Anne Scharf, A. Code for MoveApps: Make analysis for wildlife conservation a community effort. *Methods in Ecology and Evolution*. In review. Audience: wildlife researchers.

Matias, M.T., Ramage, J.M., Brodzik, M.J., Gurarie, E. 2024. Snowmelt onset and caribou (*Rangifer tarandus*) migration. *Remote Sensing* 16(13), 2391. doi: 10.3390/rs16132391. NSF funding not acknowledged. Audience: caribou researchers, managers, and stakeholders.

Orndahl, K.M., Bentzen, T.W., Berner, L.T., Ehlers, L.P.W., Hebblewhite, M., Herriges, J.D., Joly, K., Macander, M.J., Palm, E.C., Sutor, M.J., Goetz, S.J., 2024. Shifting and expanding ranges of a sub-Arctic caribou herd and associated changes in vegetation. *Ecological Applications*. In Revision. NSF support acknowledged. Audience: caribou researchers, managers, policymakers, and stakeholders.

Licenses: None

Other Conference Presentations / Papers:

Barbour, N., Kelly, A., and Gurarie, E. 2024. The permeability R package: A maximum-likelihood based tool to quantify the permeability of linear barriers to animal movement. International Statistical Ecology Conference. July 2024. Wales, England. Audience: Statistical ecology students, academics, and professionals.

Barbour, N., Kelly, A., and Gurarie, E. 2024. The permeability R package: A maximum-likelihood based tool to quantify the permeability of linear barriers to animal movement. The Wildlife Society Annual Meeting. October 22, 2024. Baltimore, Maryland. Audience: Wildlife ecology students, academics, and professionals.

Beaupré, C. 2023. How Western Arctic herd survival patterns influence migration patterns. Western Arctic Caribou Herd Working Group Meeting. December 2023. Anchorage, Alaska. Audience: Western Arctic Caribou Herd Working Group members, researchers, policy makers, and other stakeholders.

Beaupré, C., Kelly, A., Hodson, J., McLaren, A., Lamontagne, E., and Gurarie, E. 2024. Study areas and vital rates: how sampling shapes boreal caribou population trends. The Wildlife Society Annual Meeting. October 24, 2024. Baltimore, Maryland. Audience: Wildlife ecology students, academics, and professionals.

Beaupré, C. 2024. Boreal caribou survival: Modeling approach, preliminary results, and explanatory variables. Boreal caribou survival project workshop. June 18, 2024. Yellowknife, Northwest Territories, Canada. Audience: Caribou researchers and managers from Government of Northwest Territories, Environment Canada, and Natural Resources Canada.

Couriot, O. 2024. Movement ecology of Arctic species: A multi-faceted approach. Research seminar. University of Alaska Fairbanks. August 2024. Audience: Wildlife ecology students and faculty.

Gurarie, E. 2024. Fate of the Caribou: Movements, distributions, and co-production of knowledge. Navigating the New Arctic (NNA) Community Meeting. March 5, 2024. Washington, DC. Audience: Arctic research community.

Gurarie, E. 2023. The Fate of the Caribou Project: Collaboration, Co-production and Co-management. Convergent Arctic Research Perspectives and Education (CARPE). Visiting scholar and seminar speaker. University of New Hampshire, Durham, New Hampshire. Audience: Arctic research community, students, and academics.

Gurarie, E., Couriot, O., Beaupré, C., Perra, M., Barbour, N., and Brose, A. 2024. Fate of the Caribou research update. Government of Northwest Territories research update. Audience: Government of Northwest Territories biologists and research staff.

Gurarie, E. and Barbour, N. 2024. Using quantitative methods to inform management strategies for mobile species. Invited seminar speakers. May 2024. Universidade de Matto Grosso do Sul, Campo Grande, Brazil. Audience: Wildlife ecology students and academics.

Gurarie, E. 2024. Fate of the Caribou: Movements, memory, and co-production of knowledge. Invited seminar speaker. Max Planck Institute of Animal Behavior. June 25, 2024. Konstanz, Germany. <https://www.ab.mpg.de/events/37263/478872> . Audience: behavioral ecology students, academics, and professionals.

Gurarie, E. 2024. Fate of the Caribou: Movements, memory, and co-production of knowledge. Invited seminar speaker, Biological and Earth Sciences. University of California, San Diego. January 2024. Virtual. Audience: Earth scientists and students.

Liao, Q. 2024. Water-Crossing Behaviours of Caribou Affected by Ice Melting and Freezing During Migrations in the High Arctic Areas. Navigating the New Arctic (NNA) Community Meeting. March 5, 2024. Washington, DC. Audience: Arctic research community.

Liao, Q. 2024. Water-Crossing Behaviours of Caribou Affected by Ice Melting and Freezing During Migrations in the High Arctic Areas. Marine-Estuarine Environmental Sciences Graduate Program (MEES) Annual Colloquium. March 22, 2024. Kent Island, Maryland.

Liao, Q. 2024. Water-Crossing Behaviours of Caribou Affected by Ice Melting and Freezing During Migrations in the High Arctic Areas. Wildlife Tracking & Movement Ecology Seminar. July 15, 2024. Beijing University, Beijing, China. Audience: Wildlife ecology students and academics.

Liao, Q. 2024. Water-Crossing Behaviours of Caribou Affected by Ice Melting and Freezing During Migrations in the High Arctic Areas. Behavior, Ecology, Evolution, And Systematics (BEES) Retreat 2024. September 14, 2024. Misty Mount, Thurmont, Maryland. Audience: Wildlife ecology students and academics.

Liao, Q. 2024. Water-Crossing Behaviours of Caribou Affected by Ice Melting and Freezing During Migrations in the High Arctic Areas. The Wildlife Society Annual Meeting. October 22, 2024. Baltimore, Maryland. Audience: Wildlife ecology students, academics, and professionals.

Orndahl, K.M., Berner L.T., Macander M., Goetz, S.J. 2024. Developing the Next Generation of Aboveground Vegetation Biomass Maps for the Arctic. NSF Navigating the New Arctic Community Meeting. Poster Presentation. Washington, D.C. Audience: Arctic research community.

Orndahl, K.M., Berner L.T., Macander M., Goetz, S.J. 2023. Developing the Next Generation of Aboveground Vegetation Biomass Maps for the Arctic. American Geophysical Union Annual Meeting. Oral Presentation. San Francisco, CA. Audience: Remote sensing and spatial analysis students, academics, and professionals.

Orndahl, K.M. 2024. Caribou was the reason, and everything else happened after. Guest lecture, ENV375: Arctic Ecosystem and Climate Change course. University of Alaska Southeast. April 4, 2024. Audience: undergraduate students (~15 students).

Perra, M. Gurarie, E., and Barbour, N. 2024. Research update. Environmental Monitoring Advisory Board meeting. Audience: Board members from Tłı̨chǫ Government, Kitikmeot Inuit Association, Diavik Mine, and North Slave Métis Alliance.

Perra, M., Çoban, E., Brinkman, T., Boelman, N., Crimmins, S., Dumond, A., Roberto-Charron, A., Adamczewski, J., Liston, G., Reinking, A., Mandel, M., Couriot, O., and Gurarie, E. 2024. The power of soundscape monitoring in the Arctic. The Wildlife Society Alaska Chapter Annual Meeting. Oral presentation. Anchorage, Alaska. Audience: Wildlife researchers and academics, especially biologists with the Alaska Department of Fish and Game and U.S. Geological Survey.

Other Products:

Berner, L. T., K. M. Orndahl, M. Rose, M. Tamstorf, M. F. Arndal, H. D. Alexander, E. R. Humphreys, M.M. Loranty, S. M. Ludwig, J. Nyman, S. Juutinen, M. Aurela, K. Happonen, J. Mikola, M. C. Mack, M. R. Vankoughnett, C. M. Iversen, V. G. Salmon, D. Yang, J. Kumar, P. Grogan, R. K. Danby, N. A. Scott, J. Olofsson, M. B. Siewert, L. Deschamps, E. Lévesque, V. Maire, A. Morneault, G. Gauthier, C. Gignac, S. Boudreau, A. Gaspard, A. Kholodov, M. S. Bret-Harte, H. E. Greaves, D. Walker, F. M. Gregory, A. Michelsen, T. Kumpula, M. Villoslada, H. Yläne, M. Luoto, T. Virtanen, B. C. Forbes, N. Hölzel, H. Epstein, R. J. Heim, A. Bunn, R. M. Holmes, J. K. Y. Hung, S. M. Natali, A.-M. Virkkala, and S. J. Goetz. 2024. The Arctic plant aboveground biomass synthesis dataset. Arctic Data Center. doi:10.18739/A2QJ78081.

Brose, A., Gurarie, E., Rodgers, P., and Freeman, I. Animating Bathurst caribou movements around winter roads in Northwest Territories. Video. In development. NSF support acknowledged.

Other Publications:

Beaupré, C. Collective movement. Poem on caribou behavior. Navigating the New Arctic Nuna Zine. In review. NSF support acknowledged. Audience: Arctic research community; Residents of Arctic Alaska.

Brose, A., Gunn, A., Freeman, I., Gurarie, E., Meinert, L., Guile, A., Pellissey, J., Behrens, S., Jacobsen, P., Couriot, O., Salganek, S., and Kauffman, M. 2024. Shifting trails: the shrinking range of Bathurst Caribou. ArcGIS Storymap. <https://shiftingtrails.caribou.esf.edu>. NSF support acknowledged. Audience: Policy makers and stakeholders in Northwest Territories and Nunavut; general publics.

Brose, A. 2024. Sharing caribou knowledge: Fate of the Caribou Project. Caribou Trails newsletter. Western Arctic Herd Working Group. Summer 2024. <https://westernarcticcaribou.net/caribou-news/>. NSF support acknowledged. Audience: All residents of the Western Arctic Caribou Herd's range (approximately 10,000 people); Policy makers and other stakeholders throughout Alaska.

Brose, A., Larue, B., and Gunn, A. Caribou and grizzlies in a New Arctic: an interview with researcher Benjamin Larue. Navigating the New Arctic Nuna Zine. In review. NSF support acknowledged. Audience: Arctic research community; Residents of Arctic Alaska.

Brose, A. The way is changing. Poem on caribou migration. Navigating the New Arctic Nuna Zine. In review. NSF support acknowledged. Audience: Arctic research community; Residents of Arctic Alaska.

Brose, A., Gurarie, E., Gunn, A., Couriot, O., and Guile, A. 2024. Are Bathurst caribou overlapping more with other herds in winter? Graphical research summary. Wek'èezhii Renewable Resources Board. English and Tłıchǵ. https://www.wrrb.ca/sites/default/files/BathurstOverlap_digital.pdf. NSF support not acknowledged. Audience: Residents of the Bathurst Caribou Herd range; Policy makers and stakeholders in northern Canada.

Brose, A., Matias, M.T., Ramage, J.M., Brodzik, M.J., and Gurarie, E. Does spring snow melt trigger caribou migration? Graphical research summary. In development. Audience: Residents of caribou range in Alaska and Canada; Policy makers responsible for caribou and climate action.

Couriot, O., Beaupré, C., Perra, M. and Phelan, O. December 2023. Collaborative research on Barren-ground caribou - Report of the discussions between the North Slave Métis Alliance (NSMA) and Fate of the Caribou (FotC). Audience: Members of the North Slave Métis Alliance, research collaborators.

Couriot, O., Ware, J. and Sutor, M. April 2024. Movement and Distribution of the Porcupine Caribou Herd – Report prepared for the Government of Yukon.

Couriot, O., Ware, J. and Sutor, M. August 2024. Porcupine Caribou Herd Annual Core Range – Report prepared for the Government of Yukon.

Couriot, O., Ware, J. and Sutor, M. September 2024. Sensitive Habitats of the Porcupine Caribou Herd - Report prepared for the Government of Yukon.

Gunn, A. 2024. Genome assembly of a Canadian Arctic Island (Banks Island) muskox – encouraging news for conservation. Springer Nature Research Communities. Blog.
<https://communities.springernature.com/posts/genome-assembly-of-a-canadian-arctic-island-banks-island-muskox-encouraging-news-for-conservation>. NSF support not acknowledged. Audience: Arctic ungulate researchers.

Russell, D., and A. Gunn. 2024. Arctic Conservation Forecast project (ARCON4): assessing vulnerability of migratory tundra caribou to climate change. Report prepared for World Wildlife Fund Global Arctic Programme. NSF support not acknowledged. Audience: Policy makers responsible for Arctic policy.

Russell, D., R. White, and A. Gunn. 2024. Understanding productivity of North American migratory tundra caribou (*Rangifer tarandus*): Role of vital rates and climate. Department of Environment and Natural Resources, Manuscript Report 312, Government of Northwest Territories, Yellowknife, Northwest Territories, Canada. NSF support not acknowledged. Audience: Northwest Territories caribou researchers, land managers, and policymakers.

Patent Applications: None

Technologies or Techniques:

Barbour, N., Kelly, A., and Gurarie, E. “permeability” R package. In development.

A maximum-likelihood based tool to quantify the permeability of linear barriers to animal movement, designed to help policymakers and researchers assess the impacts of natural and anthropogenic barriers to animal movement. Developed in partnership with Government of Northwest Territories.

Beaupré, C., Couriot, O., Brose, A., Gurarie, E., Billingsley, B., Morey, J., Druckenmiller, M., Gunn, A., Zaprosyan, A., Cheshire, M., and Earls, J. Knowledge of the Caribou database prototype. In development.

A database designed to compile Traditional and Local Ecological Knowledge (TLEK) observations of caribou from existing reports and documents recorded at caribou workshops and herd-specific working group meetings. These observations provide an important local record of the past and present state of the Arctic environment and caribou, but participation and knowledge sharing in these meetings represents a significant time burden on local community members. The goal of this database is to alleviate repeated requests for information while compiling community-based observations of caribou ecology for posterity.

Gurarie, E., Couriot, O., and Brandão Biebuhr, B. Tuktutools: a package of tools for studying tuktut (caribou) spatial patterns. <https://github.com/ocouriot/TuktutTools>

The Tuktut Tools R package is a set of tools developed by the Fate of the Caribou team that allows researchers to analyze caribou spatial and behavioral patterns. To date, the package includes functions to estimate individual calving times using movement rate analysis, identify migration timing, compute sociality indices (how often individuals are interacting with each other), estimate calving and early rearing areas, delineate overall seasonal and daily ranges, and partition behavior into transient and residential stages using GPS collar data. The package also includes tools to map and animate location data to visualize animal movements more easily. Efforts this year were

focused on debugging and fixing issues identified by researchers who are actively using it. To date, we know researchers at Government of Northwest Territories and elsewhere are using the package.

Gurarie, E. 2024. Tuktumigration: an R package for estimating individual and collective migration behavior. <https://github.com/EliGurarie/Tuktumigration>

An R package for estimating collective migration behavior the hierarchical migration model developed in [Gurarie et al. \(2019\)](#) to estimate migration timing for barren-ground caribou across North America. This package uses a Bayesian MCMC sampler using [RStan](#). To date, we know researchers at Government of Northwest Territories and elsewhere are using the package.

Thesis / Dissertations: None

Websites or Other Internet Sites:

Project homepage: <https://FateOfTheCaribou.esf.edu>
Storymap on Bathurst caribou herd range shifts: shiftingtrails.caribou.esf.edu
Project Facebook page: <https://www.facebook.com/FateOfCaribou>
Tuktumigration repository: <https://github.com/ocouriot/Tuktumigration>
Tuktumigration repository: <https://github.com/EliGurarie/Tuktumigration>
Arctic Plant Aboveground Biomass Synthesis Dataset:
<https://arcticdata.io/catalog/view/doi%3A10.18739%2FA2QJ78081>

PARTICIPANTS

What individuals have worked on the project?

Name	Most senior project role	Nearest person month worked	Describe contribution
Eliezer Gurarie	PI	6	Led the Fate of the Caribou team, organized PI meetings. Co-authored several papers and posters. Presented results and research progress at various forums. Developed analytical tools presented in the Tuktumigration, Tuktumigration, and Permeability R packages. Completed several analyses of caribou sociality and habitat use and led collaboration with Indigenous and local partners in the Arctic.
William Fagan	Co-PI	1	Organized research by graduate and undergraduate students and mentored them Contributed to design, analysis, and writing for journal manuscripts
Anne Gunn	Co-PI	2	Provided subject matter expertise during development of project research questions

			and outreach materials. Reviewed research proposals and manuscripts for post-doctorate and graduates. Co-authored outreach materials. Served on the Wek'èezhìi Renewable Resources Board technical committee and attended Technical Advisory Committee meetings for mining operations on behalf of the Kivalliq Inuit Association.
Scott Goetz	Co-PI	1	Participated in project planning meetings, guided remote sensing work
Logan Berner	Co-PI	2	Project management & mentorship. Led tundra biomass field data synthesis, manuscript, and dataset.
Katie Orndahl	Post-doc	3	Led pan-Arctic tundra biomass mapping, prepared manuscript, conducted outreach.
Skye Salganek	PhD Student	12	Developed habitat models for Western Arctic Caribou Herd.
Mark Hebblewhite	Co-PI	1	Project management & mentorship. Co-led camera-trapping project design, HMM methods paper, and revision of Ehler et al. papers.
Benjamin Larue	Post-doc	12	Co-led study on camera-trapping for predator-prey dynamics on calving grounds. Co-led project on caribou behavioral segmentation using Hidden Markov Models (HMMs). For both projects, conducted analyses and prepared manuscript. Started new partnerships for continental-wide caribou road ecology study.
Libby Ehlers	Post-doc	1	Co-led revisions on two manuscripts. Led data sharing paper for Fortymile caribou video-camera data sharing and banking.
Ophélie Couriot	Post doc	12	Met with collaborators, analyzed data, wrote papers, communicated results through oral presentations, co-mentored PhD students, built and disseminated analysis tools
Marron McConnell	PhD Student	5	Collaborated with representatives from the Government of Northwest Territories (GNWT) to draft a new scope of work to better align with GNWT requirements for data use and signed a new data sharing agreement. Created new analysis framework and model structure to explore demography from a more theoretical standpoint in keeping with the new scope of work. Wrote extensive new code to implement and analyze demographic

			models, explore parameter space, classify model outcomes, etc. In-progress: finalizing model results, conducting sensitivity analysis, beginning to summarize conclusions
Qianru Liao	PhD Student	5	Recently passed her proposal defense and officially became a PhD candidate. Continued research examining the relationship between caribou water crossing behavior and the freezing and thawing of Contwoyto Lake using GPS tracking data from the Bathurst and Beverley herds within 50 kilometers of the lake. Mapped the spatial patterns of ice-breaking and freezing events and categorized caribou crossing behavior into three groups: Crossing, Circumventing, and Unknown Events. To refine the classification of Unknown Events, she trained a Cost-Sensitive Random Forest model, achieving 92% accuracy. The most influential factors in the model's success were speed ratios (circumvent vs. reference speed), logarithmic speed ratios (direct vs. reference speed), and albedo along potential crossing paths. In the coming month, further analyses of the Unknown Crossing Events during the Fall migration will be completed and a manuscript will be drafted on these findings.
Stephanie Chia	PhD candidate	2	Computational assistance with animal movement data
Gayatri Anand	PhD student	2	Computational assistance with animal movement data
Frank McBride	PhD student	1	Computational assistance with animal movement data and mathematical models of animal movement
Jessica Morey	Undergraduate	1	Developed and helped populate a database on scientific and cultural information pertaining to the migration of caribou in northern North America
Katherine Tootchen	Undergraduate	1	Computational analyses of trail establishment and trail following by animals in polar environments
Megan Perra	PhD Student	12	Developed animal-borne acoustic recording device. Deployed acoustic recorders with

			collaborators. Presented research findings in several venues. Mentored undergraduate research associates.
Nicole Barbour	Post-doc	7	Developed code to estimate permeability of linear barriers to caribou.
Chloe Beaupré	PhD Student	12	Research on sociality and migration memory of caribou, effects of climate change on migration and demographics. Presented research findings in several venues. Traveled to partnering communities and conducted field work with collaborators. Lead development of the Knowledge of the Caribou database. Mentored undergraduate research associates.
Anna Brose	Other Professional	6	Communications and program management. Lead efforts to generate and design outreach materials (storymap, graphical research summaries, etc.), administrative support, proofreading publications, and writing project outreach and reporting materials. Assisted in development of Knowledge of the Caribou database.
Megan Cheshire	Undergraduate	1	Consulted on development of Knowledge of the Caribou database. Reviewed documents and entered observations and knowledge about caribou into a test database. Participated in planning meetings.
Jaylen Earls	Undergraduate	1	Consulted on development of Knowledge of the Caribou database. Reviewed documents and entered observations and knowledge about caribou into a test database. Participated in planning meetings.
Brendan Billingsley	Contractor	1	Led software development on Knowledge of the Caribou database, mentored undergraduate computer science intern

Has there been a change in the active other support of the PI/PD(s) since the last reporting period?

Yes No

If Principal Investigators (PIs)/Project Directors (PDs) and co-PIs/co-PDs select “Yes,” they are required to upload their most up-to-date Current and Pending Support document in an NSF-approved format to notify NSF that active other support has changed since the award was made or since the most recent annual report. To access the NSF-Approved formats, you can find them here: <https://www.nsf.gov/bfa/dias/policy/cps.jsp>

What other organizations have been involved as partners?

Global Initiative on Ungulate Migration

Research group
Laramie, WY
Consultation and production of maps for the Bathurst Caribou storymap project

Wek'èezhìi Renewable Resources Board
Governmental
Yellowknife, NWT, Canada
Consultation on Bathurst Caribou storymap project and in identifying research priorities

National Park Service
Governmental
Denver, Colorado
Data sharing and research co-authorship

North Slave Métis Alliance

Exchange for Local Observations and Knowledge of the Arctic

Tłıchǫ Government

Kugluktuk Angoniatit Association, Kugluktuk, NU, Canada. Provided data and expertise on caribou predator prey dynamics, and personnel for fieldwork.

Kivalliq Inuit Association

Government of the Northwest Territories Environment and Resources

Cooperative Institute for Research in the Atmosphere, Colorado State University. Fort Collins, Colorado.
Provided snow data and expertise in snow – wildlife ecology.

Have other collaborators or contacts been involved?

Yes

No

Wyoming Migration Initiative, Government of Yukon, Alaska Department of Fish and Game, Bureau of Land Management, Yukon Department of Environment, Environmental Climate Change Canada, WWF Canada

IMPACTS

What is the impact on the development of the principal discipline(s) of the project?

We are combining cross-discipline technologies to develop new methods of studying wildlife, in addition to contributing to the understanding of caribou ecology. PhD student Perra is developing and testing novel animal-borne acoustic recording devices to directly measure the soundscape caribou experience. This has involved building and programming special drop-off devices (hardware that allows the collar to release off the animal at a given time). She is also using machine learning to process the thousands of hours of acoustic recordings. Co-PI Couriot, PI Gurarie, Post-doc Barbour, and PhD student Beaupré are programming new analytical tools that use rate-change cues in animal movement data to detect life

history events like calving, insect harassment, and social aggregation. These tools and others we are developing are already being used by other researchers and natural resource managers to inform policy decisions.

We developed novel methods to map tundra cover, height, and biomass using field surveys, uncrewed aerial vehicles, and satellite remote sensing. We also established a standard method for synthesizing and harmonizing plant and lichen biomass data from a variety of data collection techniques, leading to the Arctic Plant Aboveground Biomass Synthesis Dataset, which can be used for many applications in Arctic ecology research. PI Berner led this effort, which brought together 58 researchers from 30 institutions who together contributed field measurements from 2,327 sample plots at 636 field sites in seven countries. This was a huge community effort and yielded the largest-ever collection of harmonized tundra biomass field measurements. This dataset was described in a peer-reviewed paper (Berner et al. 2024a) and publicly archived with the Arctic Data Center (Berner et al. 2024b) for community use. Our efforts to map changes in plant functional groups across Alaska, Canada, and the broader Arctic have produced several large-scale datasets and, when coupled with caribou tracking data, are answering long-standing questions about the drivers of caribou-habitat relationships.

Post-doc Ehlers's work investigating spatial foraging and juvenile survival of the Fortymile Caribou Herd produced important results for caribou population management and caribou ecology more generally. Using state-of-the-art GPS video collars that, for the first time, allowed fine-scale data collection of behavior and food choice in wild caribou, we found that density-dependence theory on foraging behavior scales down from the habitat selection all the way to the bite. This research highlighted a spatial foraging tradeoff by caribou between shrubs and lichen in summer. Our results show evidence of increased resilience of shrubs to density-dependent effects of caribou grazing and highlight potential risks of overgrazing to sensitive species, like lichens, in a changing global climate. Neonate survival increased in areas with earlier snowmelt, supporting the hypothesis that the availability of early spring vegetation affects juvenile survival ("bottom-up effects"). In our soon-to-be-submitted methods article, we show that Hidden Markov Models are extremely sensitive to temporal scale and urge caution when using them. Hidden Markov Models are an increasingly popular method in wildlife ecology, so our sensitivity analysis could have far-reaching impacts. Post-doc Larue's research on predator-caribou dynamics on the Bathurst calving grounds has revealed significant spatiotemporal overlap of grizzly bears and caribou, especially at peak calving. This result is important to inform caribou predator management, which historically has focused on wolves. As climate change alters the distribution of predators and caribou alike, our result will inform a broader view on predator-caribou management.

We have many ongoing analyses that will provide insight into caribou ecology, demography, and responses to human infrastructure and climate change. Our analysis of caribou survival relative to migratory behavior, caribou response to roads, reproductive success relative to environmental factors and calving location, effects of changes to Arctic vegetation regimes, and impacts of freeze-thaw events to caribou water crossings during migration, to name a few, will increase understanding of the complex ways caribou, climate, and people interact. Our ongoing research is geared toward not only improving the scientific understanding of caribou, but producing results that on-the-ground caribou stewards can use to prioritize habitat protections, determine harvest quotas, and improve policy.

What is the impact on other disciplines?

Nothing to Report

What is the impact on the development of human resources?

Our project is now fully staffed, including four post-docs, five PhD students/candidates, a program and communications manager, and several undergraduate research associates. The three undergraduate research associates at SUNY-ESF gained experience with complex datasets and co-production of knowledge while participating in development of the Knowledge of the Caribou database project. This experience has positively impacted their careers as wildlife ecology majors. At the University of Maryland, two PhD students, both women, achieved doctoral candidacy status based on their efforts to date and research plans for the remaining period of the grant; one undergraduate, a woman, has gained experience working at the social-environmental interface through her efforts to develop a database on scientific and cultural information pertaining to the migration of caribou in northern North America; and one undergraduate, a woman, has gained experience in computational modeling of animal movement, focusing on trail establishment and trail following in polar environments.

Our work has helped mentor our post-docs, PhD students, program manager, and undergraduate research associates in the use of statistical methods for spatial ecology and mapping and helped them gain experience working with diverse partners in Academia, NGOs, the government, and Indigenous communities. Our team members at all levels continue to benefit from knowledge sharing amongst our interdisciplinary group, and from our research partners.

What is the impact on teaching and educational experiences?

Several team members participated in and taught at technical workshops this year. Gurarie and Larue attended the AniMove workshop in Radolfzell, Germany. AniMove is an intensive international workshop to advance skills and methods in animal movement spatial analysis and remote data extraction and analysis. PI Gurarie was an instructor for the workshop, and led the session on behavioral segmentation (~26 participants). Gurarie and Barbour also led a short course on animal movement analysis at the Universidade de Matto Grosso do Sul in Cambo Grande, Brazil (~20 participants). Both courses were attended by students, academics, and non-governmental organization employees. PhD candidate Liao participated in the Animal Movement Analysis for Ecologists Workshop at Beijing University in Beijing, China, and taught the section on “Selecting, obtaining, and manipulating remote sensing data”.

In addition to the regular faculty teaching responsibilities of several of the project’s Primary Investigators, several of our Post-docs and PhD students have had the opportunity to teach lectures or participate as teaching assistants in their fields of study.

What is the impact on physical resources that form infrastructure?

PhD student Perra is developing tracking collars that will carry acoustic recorders to measure the soundscape caribou experience. This has involved building and programming drop-off devices (mechanisms that allow the collar to fall off the animal after a set amount of time) and developing an acoustic recording device that has enough memory but a small enough battery to collect a sufficient sample size while not being cumbersome to the animal. She continues testing her acoustic collar prototypes on captive research animals in partnership with the University of Alaska Fairbanks Large Animal Research Station, and initial deployments on free-ranging caribou in partnership with Government of Northwest Territories. Once developed, these collars will allow researchers to gather extensive data on the daily experiences and behaviors of large ungulates. While collars have been used for several decades to track animal movements across the landscape, pairing that movement data with acoustic information will provide unprecedented and fine-scale insight into the drivers behind observed behaviors.

What is the impact on institutional resources that form infrastructure?

None to report

What is the impact on information resources that form infrastructure?

We have published several papers and articles detailing the methods and results of our research to date, which will inform future research and caribou management decisions. Development of the storymap narrative of the Bathurst Caribou Herd, graphical research summaries, and animation of caribou movements around mining roads will also be useful tools for caribou stewards to illustrate the plight of the herd and its impacts on local communities.

Our team at Northern Arizona University published a huge synthesis dataset on Arctic plant aboveground biomass, available through the Arctic Data Center. The dataset significantly increases the scale and resolution of vegetation estimates across seven Arctic countries. The potential applications for this dataset in ecological research is immeasurable.

We are working with the Exchange for Local Observations and Knowledge of the Arctic to build a searchable database of traditional knowledge and local observations of caribou from Indigenous knowledge keepers from existing meeting notes and public records. This database will increase access to knowledge while alleviating the time demands placed on knowledge keepers who are often asked repeatedly to share their knowledge with outside researchers.

Between all our team members, we have several manuscripts in review, revision, or press. These papers provide not only ecological insight, but present new or improved techniques for analyzing data on highly-mobile animals and their habitat.

What is the impact on technology transfer?

This year we continued support and improvement on the TuktuTools and LandsatTS packages for R, which were both published last year. R is an open-source coding language that allows for free access to software packages, encouraging collaboration and innovation. Our TuktuTools and LandsatTS packages are publicly available, free to use, and continuously growing. We are also developing two additional packages (TuktuMigration and Permeability), to increase wildlife managers' access to ecologically sound and effective tools for estimating animal movement under various conditions.

What is the impact on society beyond science and technology?

Our ongoing partnerships and outreach efforts with caribou co-management boards and state and territorial governments have directed our research questions to ensure we are producing actionable results that are relevant to the local communities that rely on caribou. We are also using our technical skills to produce communication tools for caribou stewards to share the multi-faceted challenges caribou face with their communities and policy makers. For example, PI Gurarie and Program manager Brose are using Gurarie's skills with movement data and Brose's knowledge of graphic design and the Adobe suite to build a detailed animation of caribou response to mining roads. Indigenous people have been telling policy makers for years that mining infrastructure impacts caribou; now they will be able to demonstrate that visually.

One of the research directions we have taken that has been of greatest interest to our partners in the North is the analysis of potential impacts of road infrastructure on caribou populations, and the fragmentation of large, overlapping herds in light of proposed all-season roads in the Arctic. PI Gurarie and Post-doctoral researcher Barbour are addressing these issues, combining our expertise on behavioral and ecological modeling to parameterize and explore various scenarios of road infrastructure, and – together with Northern partners who have regulatory authority over construction – to inform design measures that might mitigate the effects of the roads.

Caribou are critical to the culture, lifeways, food security, and well-being of Indigenous and non-Indigenous peoples of the Arctic. Our research is already helping local communities and caribou management boards make more informed decisions about the harvest and stewardship of caribou and their habitat. As we move forward, our primary goal is to continue producing results and tools that help inform on-the-ground caribou management.

What percentage of the award's budget was spent in a foreign country?

6.5% of total budget spent on travel and supplies while in Canada.

CHANGES / PROBLEMS

Changes in approach and reason for change: In December 2023, a disappointing and unexpected conflict arose regarding use of data in the data-driven modeling framework which negatively impacted the overall research program of the University of Maryland subaward and PhD student Marron McConnell in particular. In particular, the primary data owners and their partners withdrew support (regarding data-sharing and research collaboration) for the proposed demographic modeling, citing overlap in research efforts. The intended research plan regarding the caribou demographic modeling component of the Fate of the Caribou grant, as outlined in the original grant proposal, was rendered unfeasible and was thus abandoned. Much of the progress from the prior year was lost and the entire research program had to be re-imagined. As a result, the primary research questions, methodology, and goals of McConnell's portion of the project were replaced by a theory-only model.

PhD Student Skye Salganek is starting a permanent, full-time job with the National Park Service in fall 2024. She will remain in the PhD program at Northern Arizona University but will transition off this project and instead focus on wildlife in the Southwest USA. Nevertheless, she plans to finish the analysis and paper describing the Western Arctic Caribou Herd winter resource selection as her first dissertation chapter.

Actual or Anticipated problems or delays and actions or plans to resolve them: Nothing to report

Changes that have a significant impact on expenditures: Post-doc Couriot is being hired as an assistant professor at the University of Alaska Fairbanks Department of Biology and Wildlife. She will still be a collaborator and faculty partner on this project but will no longer be paid through this grant beginning in the spring of 2025.

Significant changes in use or care of human subjects: Per the policy of the State University of New York, PI Gurarie, post-doc Couriot, and graduate students Perra and Beaupré completed training on conducting research with human subjects prior to facilitating focus groups with members of the North Slave Métis Alliance.

Significant changes in use or care of vertebrate subjects: Nothing to report

Significant changes in use or care of biohazards: Nothing to report

Has there been a change in your primary performance site location from the originally proposed? If so, please provide the location of your new primary performance site and reason for the change in location.

Nothing to report